Source: docs\PAYMENT_LIVE_CUTOVER.md | Export: 11 November 2026

# Payment queue cutover preparation

These changes prepare a cutover; they do not activate LIVE or demonstrate a real LIVE transaction. Keep environment variables and Razorpay Dashboard settings unchanged until the owner approves the release and cutover.

## Worker indexes and deployment dependency

The worker queries **both** mode and key ID before applying each 20-record limit. The due-time ordering and existing time budget remain. Records belonging to TEST or a rotated key are preserved and cannot consume the LIVE batch. Their original credentials remain necessary for any unresolved reconciliation.

Target: `parinaya-silks`, database `(default)`, query scope `COLLECTION`.

| Collection | Ordered fields |
|---|---|
| `paymentEvents` | `mode ASCENDING, keyId ASCENDING, nextCheckAt ASCENDING` |
| `paymentAttempts` | `mode ASCENDING, keyId ASCENDING, nextCheckAt ASCENDING` |

The manifest omits the implicit final `__name__ ASCENDING`, matching repository convention. Both missing definitions were subsequently created through the owner's approved create-only procedure on 9 November 2026. Existing indexes were preserved; there is exactly one matching definition per collection. Live metadata now confirms:

| Collection | Production index ID | State |
|---|---|---|
| `paymentEvents` | `CICAgLiI_5oK` | READY |
| `paymentAttempts` | `CICAgPig2YMK` | READY |

Matching read-only queries using the current worker's TEST mode/key binding both succeeded with zero due records. The initial query immediately after READY briefly returned code 9; subsequent checks succeeded after index availability propagated. Do not recreate these indexes or publish the entire manifest during app release. For another environment, wait for matching READY indexes before deploying this worker. Emulator success alone does not establish production index readiness. The worker deliberately fails rather than falling back to an unfiltered queue if an index is missing. No security rule change is required.

## Retiring terminal TEST scheduling

`scripts/retire-test-payment-schedules.ts` defaults to a read-only preview. It targets only `parinaya-silks/(default)`, scans at most 1,000 TEST attempts in pages of 100, and logs aggregate counts only. It uses the existing server credential loader; point `GOOGLE_APPLICATION_CREDENTIALS` to the trusted local key file or use the existing secure server credential variables. Never paste credentials.

```
node --import tsx scripts/retire-test-payment-schedules.ts --project=parinaya-silks '--database=(default)'
# After reviewing the current candidate count N:
node --import tsx scripts/retire-test-payment-schedules.ts --project=parinaya-silks '--database=(default)' --apply --expected-count=N
```

Apply refuses a count mismatch before writing. Each candidate is then reread with its source order inside an independently retryable transaction. Only `paymentAttempts/{id}.nextCheckAt` is deleted, only for a known matching TEST provider mapping with an exact captured amount, consumed reservation and no refund/review evidence. Unknown, active, released, LIVE, refunded and review cases remain scheduled. No documents are deleted. All attempt evidence, unrelated fields, source orders, history and webhook events are preserved.

On interruption, rerun preview to obtain the remaining count and resume. Concurrent changes to an order/attempt cause transaction retries or a reported skip. An already retired attempt is a no-op. This is schedule retirement, not an archival operation and not a way to clear an unresolved payment.

New settlement logic also removes periodic scheduling for clean TEST captures. It keeps daily LIVE capture checks, active recovery, refund/review polling and explicit status/admin/webhook reconciliation. A later refund observation restores scheduling. The old deployed code can reschedule a TEST capture if it reconciles it again; recheck after the approved application release. Do not disable the existing recovery worker during preparation.

The first production preview on 9 November found **13** eligible scheduled TEST captures, rather than the 12 in the earlier audit. The owner explicitly approved all 13. Apply retired all 13 schedules with zero concurrent-change skips. Read-back comparison confirmed that only `nextCheckAt` changed: all other attempt fields, all 13 source orders and all webhook events were unchanged. A subsequent preview found zero remaining eligible schedules. No

payment provider calls were made.

## Preview protection and remaining acceptance

`VERCEL_ENV=preview` rejects payment HTTP and service operations, including initiation/resume, verification, reconciliation, webhook acceptance and worker processing. It cannot be bypassed by a production Origin, valid worker secret, an allowlisted UID or a copied LIVE activation flag. Production/local behavior and COD services are unchanged; credential scopes remain Production-only.

Before LIVE activation: resolve any old-mode pending events/active holds using their original credentials, confirm the two READY indexes, approve/release the local changes, install the approved LIVE credentials and matching seven-event webhook, explicitly activate LIVE and obtain a fresh mode/key-bound worker heartbeat. These are separate authorized operations. Then perform owner-approved hosted checks for COD and a real low-value LIVE transaction. Neither local tests nor historical TEST acceptance confirms LIVE settlement or production cutover.

## Initial implementation verification (9 November 2026)

- Current Production metadata reported READY commit `b8e13c5`, Node 24, rather than the supplied baseline `156c520`. Payment backend/entrypoint source is unchanged between those commits. These new application changes remain local.
- The initial read-only worker queries returned missing-index code 9. After the separately approved index creation, both indexes are READY and both exact queries succeed. No security rules were published.
- TypeScript and production frontend build passed; the existing chunk-size warning remains. All 192 unit tests passed, with zero skips.
- 30 focused emulator tests passed: existing payment lifecycle (17), new cutover coverage (6), and seven order/coupon/loyalty/security regressions. Mixed-mode and rotated-key queues retain full 20-event/20-attempt capacity in each mode; maintenance is dry-run/count-checked/idempotent, and Preview rejects before protected dependency access. The new suite is included in `test:admin`.
- Packaged Vercel payment Web exports passed Preview rejection, original-byte HMAC/durable acceptance/tamper rejection with loopback emulators, plus signed-out and configuration-failure checks. Vercel Node 24 artifact executed locally on Node 22; this does not claim a hosted invocation of the new code.
- No new real TEST/LIVE payment, commit, push, deployment, environment-variable change or Razorpay Dashboard change was performed. Production changes were limited to the explicitly approved scheduling-field cleanup and two indexes.

## Final review - 9 November 2026

- Reviewed all 13 changed/new paths, including application code, regression tests, maintenance tooling and documentation. No dependency, security-rule, environment, frontend business-flow or unrelated UI changes are included.
- Fresh production metadata confirmed both index IDs above remain READY with COLLECTION scope and implicit `__name__ ASCENDING`. Both exact mode/key-bound queries succeeded with zero due records.
- Read-only production reconciliation found 13 TEST attempts with 13 matching CAPTURED/consumed source orders, 43 processed TEST webhook events, zero missing source orders, zero active reservations and zero pending checkout locks. No TEST attempt has a due or future schedule. The TEST worker heartbeat was fresh (42 seconds old at the 11:57 IST observation).
- The earlier approved cleanup's before/after comparison recorded that only the 13 scheduling fields changed, with orders and webhook events unchanged. This final review performed no production writes and did not rerun cleanup.
- Passed: 192 unit tests; 40 tests across 18 backend/rules regression files; payment UI, checkout coupons, checkout context, checkout design and Bag/Wishlist browser suites; TypeScript; production build; packaged catalog and payment runtime checks; and `git diff --check`. No assertions were removed or skipped. The first checkout-design browser run reported offline while expecting its loading skeleton; the suite passed unchanged on a fresh run, including its explicit offline/retry coverage. The existing build chunk-size warning remains.
- Production is still READY at `b8e13c5`; the new application behavior has only been verified locally. The packaged Node 24 Vercel artifact ran locally on Node 22, not in a newly deployed function. Hosted acceptance of this code remains part of the proposed application release below.

## Proposed application release (requires owner confirmation)

Final review added a Preview guard at the start of direct internal `reconcile()` as well as the HTTP/customer/configuration guards. The regression checks that a direct Preview reconciliation rejects before Firestore or provider access.

Proposed commit: `fix(payments): isolate recovery queues and block Preview processing`

Stage only these 13 reviewed paths, preserving any other work:

```
docs/PAYMENTS.md
docs/PAYMENT_LIVE_CUTOVER.md
firestore.indexes.json
server/payments/config.ts
server/payments/http.ts
```

```
server/payments/queue.ts
server/payments/service.ts
server/payments/settlement.ts
scripts/admin-emulator-suite.ts
scripts/payment-cutover-emulator.test.ts
scripts/payments-runtime.test.mjs
scripts/retire-test-payment-schedules.ts
src/payments/security.test.ts
```

After confirmation, recheck the final diff and remote `main` for concurrent work, stage the allowlist, inspect the staged diff, commit with the existing authorized project identity and push `main` through the established GitHub/Vercel workflow. Do not stage `.cache`, `.env*`, credentials or generated runtime artifacts. Do not rerun maintenance or publish any indexes/rules as part of this app release.

Keep the existing TEST payment configuration and LIVE activation setting unchanged. Verify Vercel's Production deployment is READY for the exact new commit and that the approved domains point to it. Observe the existing scheduler's fresh TEST mode/key-bound heartbeat, successful filtered queries, no unexpected pending work and no recreated terminal schedules. Check existing customer/Admin views read-only and verify Preview payment endpoints return 403 without initiating a payment. New TEST/LIVE transactions require separate authorization.

If hosted checks fail, restore the previous known Production deployment (`b8e13c5` at review time) while retaining TEST configuration, all payment evidence and both indexes; investigate before any LIVE configuration change. Application rollout is separate from installing LIVE keys/webhook, changing mode/activation, obtaining the first fresh LIVE heartbeat and approving the first real LIVE transaction.